

5. The plan

Run the planner once the registration is accepted and the sensor values are in `sensor_config.json`. It writes the commanded pose of every capture, in the order to do them.

```
python3 -m sensorperf.cli.plan_stations --out plan/ --registration registration.json \ --sensor-config sensor_config.json --seed 1
```

Write the seed in `session_log.md`. The same seed and inputs give the same plan again. Without `--registration` the planner writes the target poses in the camera frame only; without `--sensor-config` it uses the indicative sensor geometry and says so loudly, because the field-of-view fit, the lateral offsets in millimeters, the expected depth quantum, and the capture budget (§13) all depend on the real values. Appendix B lists every option.

The planner writes five files in the output folder:

`poses.csv`: one row per commanded pose.

`plan_summary.txt`: poses and frames per series and station, the capture budget, every adjustment, and every warning. Read it before you start.

`plan.png`: the planned target centers, a picture of the same kind as Figure 6.

`targets.json`: the target definitions the plan was made with.

`parameters.json`: the parameter values the plan was made with (appendix A).

One geometric ladder of Z stations serves the whole procedure: 9 stations of the ladder at a ratio of $2^{1/4}$ (four per octave), 400, 476, 566, 673, 800, 951, 1131, 1345, 1600 mm. Series A, C, and D visit all of them; A also adds the 2 legacy depths 700 and 1000 mm, so that the existing metrics can be computed at the same depths as the existing data. Series B (edges) visits every second station, the 5 shape stations (400, 566, 800, 1131, 1600 mm). Series Z and the tilt sub-series of A visit every fourth station, the 3 reduced stations (400, 800, 1600 mm). The extended trials of D run at the 3 farthest stations (1131, 1345 and 1600 mm). The reference station, 800 mm (a station of the ladder), serves the warm-up check, the sentinels, the re-mount check, the field sub-series of C, the open-background variant, and the post check of D. The working range of 400 to 1600 mm carries the dagger of appendix A, and Step 4.5 confirms it.

The stations the plan visits, and the five field positions in the image, are in Figure 5. The whole plan is in Figure 6.

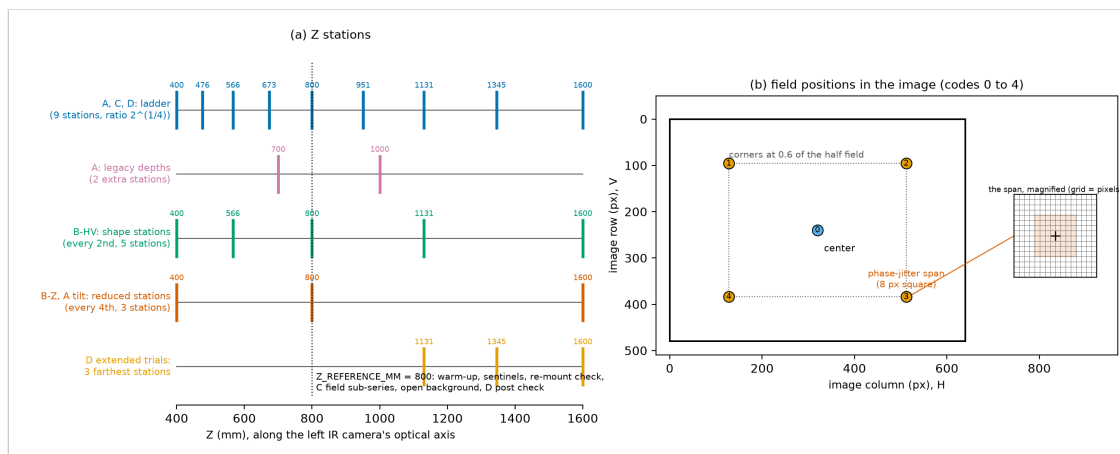


Figure 5. Left: the 9 stations of the ladder, used by A, C and D (A adds the 2 legacy depths), the 5 shape stations of B, the 3 reduced stations of Z and the tilt sub-series, and the 3 farthest stations of the extended D trials, along Z; the dotted line is the reference station. Right: the 5 field positions (codes 0 to 4) in the image, and the square span of the random lateral offsets (phase jitter) at one corner, true size and magnified.

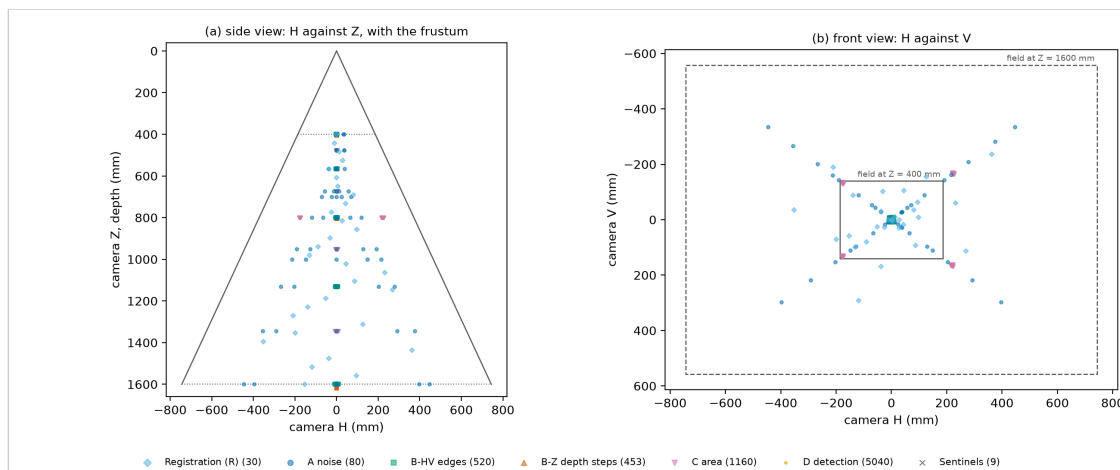


Figure 6. The default full plan: target centers in the sensor frame, side view (H against Z) and front view (H against V), colored by series, with the frustum. Each cloud around a station is a set of random lateral offsets.

What a row of poses.csv contains. Table 4 lists the procedure letters. A row has the order to execute it in (order), the identity that names the files (procedure, target_id, gap_mm, station_z_mm, field, pose_index), the number of frames (frames), a sub-series label (subseries: for example main, tilt, remount, nominal, jitter, ladder, staircase, field, extended, sentinel), the logged random seed (seed), the random lateral offset in mm at the station depth (offset_h_mm, offset_v_mm), the tilt (tilt_axis, tilt_deg), the commanded Z step and visit (step_mm, visit, series Z only), the feature index (level_index, usually empty, because one frame sees every feature of a plate), and the wanted target

pose in the camera frame (target_x_mm to target_rz_deg: position in mm and a rotation vector in degrees). With a registration the row also holds the flange pose to command in the robot base frame: position and rotation vector (base_x_mm to base_rz_deg), the rotation matrix (r00 to r22, row by row), and the quaternion (quat_w, quat_x, quat_y, quat_z). Use whichever form your robot program accepts. A last column (notes) holds extra detail as JSON.

Letter	Series	Target	See
R	Registration	T2	§4
A	Noise plate	T2	§6
B	Edges	T3a, T3b	§7
Z	Depth steps (B-Z)	T2	§8
C	Disk and cutout areas	T4, T5	§9
D	Detection trials	T4, T5	§10
S	Drift sentinels	T2	§6, step 3

Table 4. Procedure letters of the plan and of the file names.

The planner checks that each target fits the field of view at its station. A target that would not fit, for example T2 off-axis at Z_MIN, is pulled inward along its field direction until it fits. plan_summary.txt lists every such adjustment and every target that does not fit even when centered. Do not override these. Tell the engineer about any plate that does not fit when centered. The disk and cutout plates T4 and T5 are built to fit the field of view at Z_MIN with room for the random offsets; if the target set cannot be built that way, the planner stops with an error that names the plate, and the engineer changes the target design before anything is fabricated.

Robot program outline, for each row of poses.csv, in order:

Move to the pose. Use a joint move to a point short of it along the target normal, then a linear move onto it, so the approach is the same every time. For every visit of series Z the point short of the pose lies below it: back off by 2 mm toward smaller Z, then move up onto the pose, so that every visit of the series is approached from below and backlash does not enter the difference between visits (§8).

Wait the settle time of Step 4.4 (2 s unless the check raised it).

Trigger the capture of frames frames. Name the files <proc>_<target>_G<gap>_Z<zzzz>_F<field>_P<pose>_f<frame>.mc from the row: for example C_T5_G15_Z0800_F0_P017_f03.mc is procedure C (area), target T5, gap 15 mm, station Z = 800 mm, field position 0 (center), pose 17, frame 3. A target without a back plate (T2) writes G0. The frame number runs from f00.

Read the robot's actual reported (encoder-derived) flange pose, not the commanded one, and append it to the pose log (§11). Add the sensor and air temperature and a timestamp.

Move on.

Sentinel rows (letter S) appear in the plan every 60 minutes, by the planner's estimate of the clock. A sentinel is T2 at the center at $Z = 800$ mm, 30 frames. It needs T2 on the robot. If T2 is not mounted when a sentinel row comes up, ask the engineer whether to swap targets now or to do the sentinel at the next target change, and write the actual time of the capture in the pose log. After the sentinel, mount the target of the series again and do the mount check of Step 4.8.

Do the series in the order of the plan: A, then B, then Z, then C, then D. Do not move the sensor between them. Tell the engineer at once if a series stops early; the later series depend on the earlier ones (Figure 1).